

Surname <u>Memo</u> Van _____ The undersigned hereby confirms that they are familiar with and accepts the examination regulations	Initials _____ Voorletters _____ Stud. No / Nr. _____ Ondergetekende bevestig hiermee dat hulle bewus is van die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp
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Handtekening _____ Signature _____ Date _____ Datum _____

Class Test 1

2 March 2012
FSK116

Klastoets 1

Total/Totaal: 30 max

Circle: Circumference = $2\pi r$ = omtrek Area = πr^2	$\vec{a} \cdot \vec{b} = ab \cos \phi$ $ \vec{a} \times \vec{b} = ab \sin \phi$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Sphere: Area = $4\pi r^2$ Volume = $\frac{4}{3}\pi r^3$	$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$ $\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}$	$\sin^2 \theta + \cos^2 \theta = 1$ $\sin 2\theta = 2 \sin \theta \cos \theta$ $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$
Cylinder / Silinder Area = $2(\pi r^2) + 2\pi r h$ Volume = $\pi r^2 h$		
Trapezium: Area = $\frac{1}{2}(a+b)h$		

Question 1/Vraag 1

Determine the following (to 3 significant figures):

Bereken die volgende (tot 3 beduidende syfers):

$4^{2.3}$	24.25=24.3
$\log(35.39)$	1.55
$\ln(0.611)$	-0.493
111° in radians / in radiale	1.94 radians (0.617 π radians)
3.5π in degrees / in grade	630°
$\sin -56.78^\circ$	-0.837
$\tan^{-1}(3.234)$	72.8° or 1.271 rad

[7]

Half a mark was deducted for incorrect number of significant figures.

Question 2 / Vraag 2

A cylindrical pipe has an outer diameter of 30 mm and an inner diameter of 25 mm.

'n Leë silindriese pyp het 'n buitendiameter van 30 mm en 'n binne diameter van 25 mm.

(a) Determine the volume of material used to make up 2 m of the pipe. Give your final answer in m^3 .

Bereken die materiaalvolume van 2 m van dié pyp. Gee jou finale antwoord in m^3 .

[2]

$V = (\pi r_o^2 - \pi r_i^2) \times l = \pi (r_o^2 - r_i^2) \times l$ $= \pi \left(\left(\frac{30}{2} \times 10^{-3} \text{ m} \right)^2 - \left(\frac{25}{2} \times 10^{-3} \text{ m} \right)^2 \right) \times 2 \text{ m}$ $= 4.32 \times 10^{-4} \text{ m}^3$	
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(b) How much water will 2 m of this pipe contain if one end is closed?

Hoeveel water sal 2 m van hierdie pyp bevat as dit aan die een kant toegemaak word?

[2]

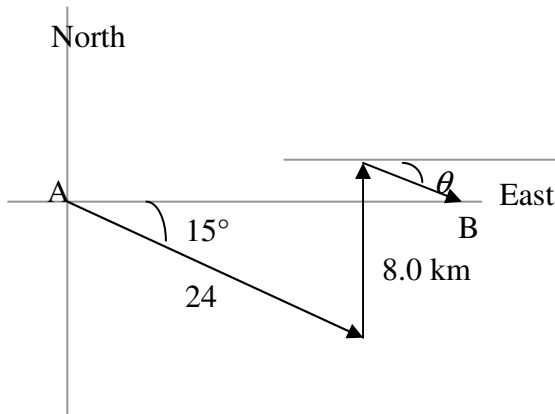
$V = (\pi r_i^2) \times l = \pi (r_i^2) \times l$ $= \pi \left(\frac{25}{2} \times 10^{-3} \text{ m} \right)^2 \times 2 \text{ m}$ $= 9.817 \times 10^{-4} \text{ m}^3$	
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Question 3 / Vraag 3

Oasis B is 25 km due east of oasis A. Starting from oasis A, a camel walks 24 km in a direction 15° south of east and then walks 8.0 km due north. Determine the distance and the direction in which the camel must walk to **now** get to oasis B.

*Oase B is 25 km oos van oase A. 'n Kameel stap van oase A, in 'n rigting 15° suid van oos vir 24 km, en dan loop hy 8.0 km noordwaarts. Bepaal die afstand en die rigting wat die kameel **nou** moet loop om by oase B te kom.*

[11]



Space is provided for a diagram and marks were allocated for the diagram. If you indicated θ on the diagram this is what angle was taken for your final answer. If no angle was shown, and no explanation was given – this answer is actually meaningless.

First displacement

$$x_1 = d_1 \cos \theta = 24 \text{ km} \cos 15^\circ = 23.18 \text{ km east}$$

$$y_1 = d_1 \sin \theta = 24 \text{ km} \sin 15^\circ = 6.21 \text{ km south}$$

Second displacement in unit vector form

$$x_2 = 0 \text{ km east}$$

$$y_2 = 8.0 \text{ km north}$$

Sum of displacements in unit vector form

$$x_t = x_1 + x_2 = 23.18 \text{ km} + 0.0 \text{ km} = 23.18 \text{ km east}$$

$$y_t = y_1 + y_2 = -6.21 \text{ km} + 8.0 \text{ km} = 1.79 \text{ km north}$$

Displacement from oasis B in unit vector form

Final displacement:

$$x_f = 25 \text{ km} = 23.18 \text{ km} + x_3 \text{ km}$$

$$x_3 = 25 - 23.18 \text{ km} = 1.83 \text{ km east}$$

$$y_f = 0 = -6.21 \text{ km} + 8.0 \text{ km} + y_3$$

$$y_3 = -1.79 \text{ km} = 1.79 \text{ km south}$$

Distance from oasis B

$$d_3 = \sqrt{x_3^2 + y_3^2} = \sqrt{1.83^2 + 1.79^2} = 2.56 \text{ km}$$

Direction to oasis B from current position

$$\tan \theta = \frac{y_3}{x_3}$$

$$\theta = \tan^{-1} \left(\frac{y_3}{x_3} \right) = \tan^{-1} \left(\frac{-1.79}{1.83} \right) = 44.4^\circ \text{ south of East}$$

Question 4/Vraag 4

Two vectors are given by $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Determine the following:

Twee vektore word deur $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ en $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$ gegee. Bepaal die volgende:

(a) $\vec{a} \times \vec{b}$

[2]

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_x b_y - a_y b_x) \hat{k} \\ &= [(3.0)(4.0) - (5.0)(2.0)] \hat{k} = 2.0 \hat{k}. \end{aligned}$$

(b) In words, describe the direction of the resulting vector in part (a). Assume that the x and y axes are parallel to the edges of the page, and that we use a right hand axis system.

In woorde, beskryf die rigting van die resultante vektor in deel (a). Neem aan dat ons 'n regter-hand asse-stelsel gebruik en dat die x en y asse parallel met die bladsy se kante is.

[1]

Out of the page / uit die bladsy

(c) $\vec{a} \cdot \vec{b}$

[2]

$$\begin{aligned} \vec{a} \cdot \vec{b} &= a_x b_x + a_y b_y \\ &= (3.0)(2.0) + (5.0)(4.0) = 26. \end{aligned}$$

(d) Determine the angle between vectors $\vec{a}$ and $\vec{b}$.

Bepaal die hoek tussen vektore $\vec{a}$ en $\vec{b}$.

[4]

$$a = |\vec{a}| = \sqrt{3^2 + 5^2} = 5.8$$

$$b = |\vec{b}| = \sqrt{2^2 + 4^2} = 4.5$$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

$$26 = (5.8)(4.5) \cos \phi$$

$$\phi = \cos^{-1} \left(\frac{26}{26.1} \right) = 5.01^\circ$$

This final answer depends significantly on rounding off of numbers. If you are sure your calculation is correct when using more significant figures, please come see me after class with your paper.

THE END / DIE ENDE